



National  
Qualifications  
2022

**X857/76/12**

**Physics**  
**Paper 1 — Multiple choice**

FRIDAY, 13 MAY

9:00 AM – 9:45 AM

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**Total marks — 25**

Attempt ALL questions.

**You may use a calculator.**

Instructions for the completion of Paper 1 are given on *page 02* of your answer booklet X857/76/02.

Record your answers on the answer grid on *page 03* of your answer booklet.

Reference may be made to the data sheet on *page 02* of this question paper and to the relationships sheet X857/76/22.

Space for rough work is provided at the end of this booklet.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



## DATA SHEET

### COMMON PHYSICAL QUANTITIES

| Quantity                               | Symbol | Value                                                             | Quantity          | Symbol | Value                              |
|----------------------------------------|--------|-------------------------------------------------------------------|-------------------|--------|------------------------------------|
| Speed of light in vacuum               | $c$    | $3.00 \times 10^8 \text{ m s}^{-1}$                               | Planck's constant | $h$    | $6.63 \times 10^{-34} \text{ J s}$ |
| Magnitude of the charge on an electron | $e$    | $1.60 \times 10^{-19} \text{ C}$                                  | Mass of electron  | $m_e$  | $9.11 \times 10^{-31} \text{ kg}$  |
| Universal Constant of Gravitation      | $G$    | $6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ | Mass of neutron   | $m_n$  | $1.675 \times 10^{-27} \text{ kg}$ |
| Gravitational acceleration on Earth    | $g$    | $9.8 \text{ m s}^{-2}$                                            | Mass of proton    | $m_p$  | $1.673 \times 10^{-27} \text{ kg}$ |
| Hubble's constant                      | $H_0$  | $2.3 \times 10^{-18} \text{ s}^{-1}$                              |                   |        |                                    |

### REFRACTIVE INDICES

The refractive indices refer to sodium light of wavelength 589 nm and to substances at a temperature of 273 K.

| Substance   | Refractive index | Substance | Refractive index |
|-------------|------------------|-----------|------------------|
| Diamond     | 2.42             | Water     | 1.33             |
| Crown glass | 1.50             | Air       | 1.00             |

### SPECTRAL LINES

| Element  | Wavelength (nm) | Colour      | Element        | Wavelength (nm)    | Colour   |
|----------|-----------------|-------------|----------------|--------------------|----------|
| Hydrogen | 656             | Red         | Cadmium        | 644                | Red      |
|          | 486             | Blue-green  |                | 509                | Green    |
|          | 434             | Blue-violet |                | 480                | Blue     |
|          | 410             | Violet      | Lasers         |                    |          |
|          | 397             | Ultraviolet | Element        | Wavelength (nm)    | Colour   |
|          | 389             | Ultraviolet | Carbon dioxide | 9550 }<br>10 590 } | Infrared |
| Sodium   | 589             | Yellow      | Helium-neon    | 633                | Red      |

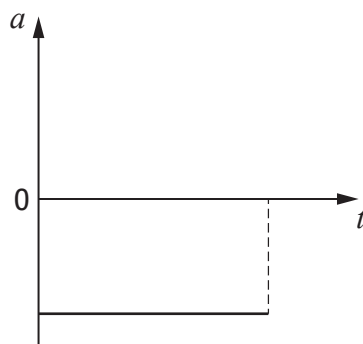
### PROPERTIES OF SELECTED MATERIALS

| Substance | Density ( $\text{kg m}^{-3}$ ) | Melting point (K) | Boiling point (K) |
|-----------|--------------------------------|-------------------|-------------------|
| Aluminium | $2.70 \times 10^3$             | 933               | 2623              |
| Copper    | $8.96 \times 10^3$             | 1357              | 2853              |
| Ice       | $9.20 \times 10^2$             | 273               | ....              |
| Sea Water | $1.02 \times 10^3$             | 264               | 377               |
| Water     | $1.00 \times 10^3$             | 273               | 373               |
| Air       | 1.29                           | ....              | ....              |
| Hydrogen  | $9.0 \times 10^{-2}$           | 14                | 20                |

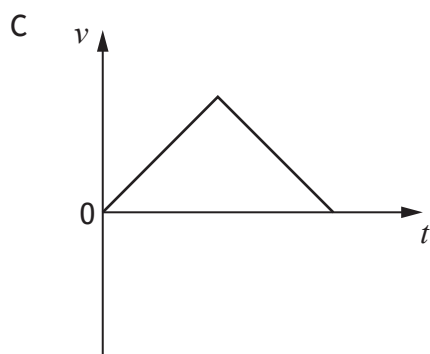
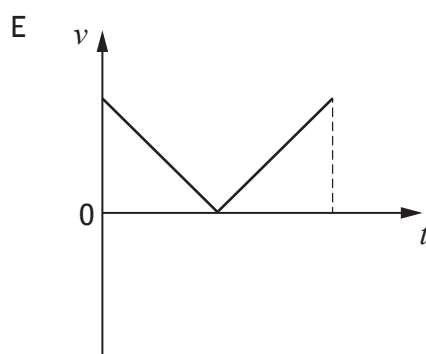
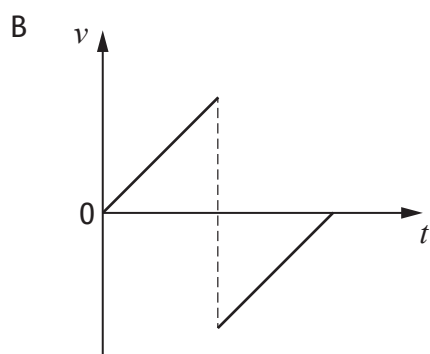
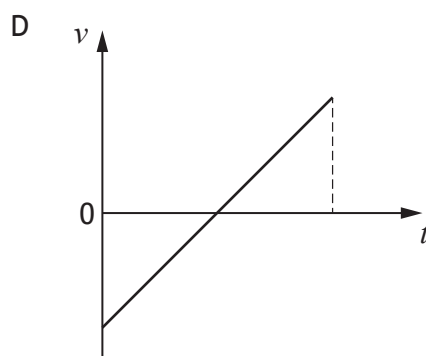
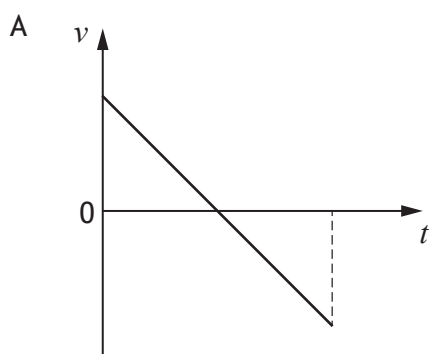
The gas densities refer to a temperature of 273 K and a pressure of  $1.01 \times 10^5 \text{ Pa}$ .

Total marks — 25  
Attempt ALL questions

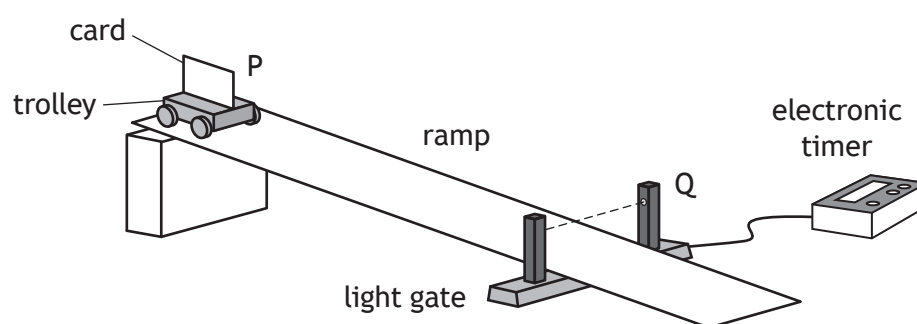
1. A ball is thrown vertically upwards and falls back to its starting position.  
The acceleration-time graph represents the motion of the ball.



Which of the following velocity-time graphs represents the same motion?



2. A student uses the apparatus shown to determine the acceleration of a trolley as it moves down a ramp.



The trolley is released from rest at point P and moves down the ramp.

A card attached to the trolley passes through a light gate at point Q.

The time for the card to pass through the light gate is displayed on the electronic timer.

The vehicle's acceleration  $a$  is determined using the relationship

$$v^2 = u^2 + 2as$$

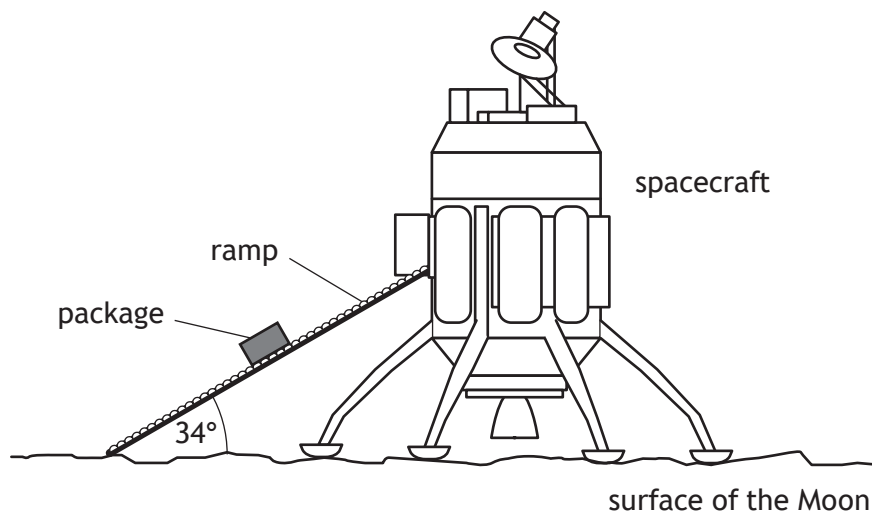
The student makes the following statements about the terms  $u$ ,  $s$ , and  $v$ :

- I  $u = 0 \text{ m s}^{-1}$
- II  $s$  = the length of the card
- III  $v = \frac{\text{distance between P and Q}}{\text{time displayed on timer}}$

Which of these statements is/are correct?

- A I only
- B II only
- C I and II only
- D I and III only
- E I, II and III

3. A spacecraft unloads cargo on the surface of the Moon.  
The gravitational field strength on the Moon is  $1.6 \text{ N kg}^{-1}$ .



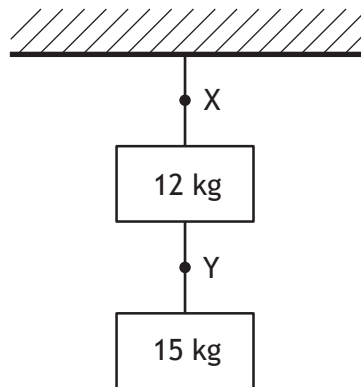
A package of mass  $3.0 \text{ kg}$  moves down the ramp.

The component of the weight of the package acting parallel to the ramp is:

- A  $0.89 \text{ N}$
- B  $2.7 \text{ N}$
- C  $4.0 \text{ N}$
- D  $4.8 \text{ N}$
- E  $16 \text{ N}$ .

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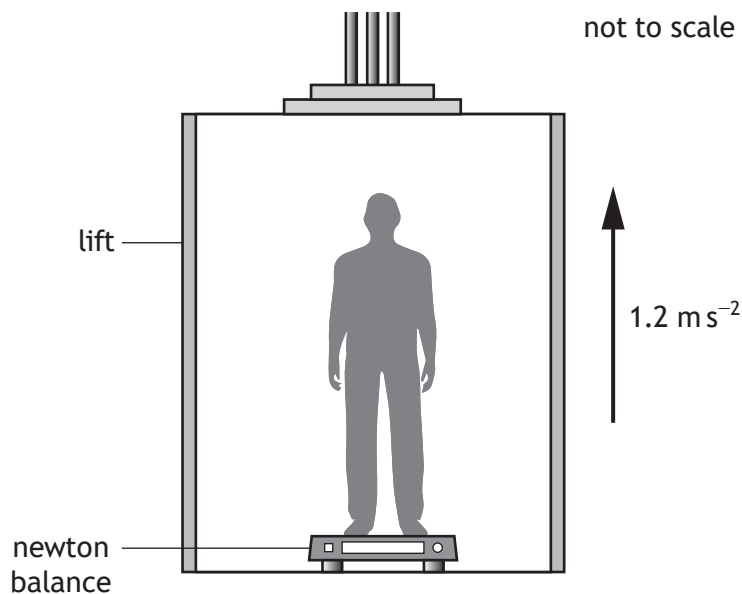
4. Two blocks are suspended from a ceiling by ropes as shown.



Which row in the table shows the tension in the rope at point X and the tension in the rope at point Y?

|   | Tension at point X<br>(N) | Tension at point Y<br>(N) |
|---|---------------------------|---------------------------|
| A | 27                        | 15                        |
| B | 120                       | 29                        |
| C | 120                       | 150                       |
| D | 260                       | 29                        |
| E | 260                       | 150                       |

5. During an experiment a student inside a lift stands on a newton balance.



The mass of the student is  $50.0 \text{ kg}$ .

The lift accelerates upwards at  $1.2 \text{ m s}^{-2}$ .

The reading on the newton balance is:

- A  $60 \text{ N}$
  - B  $430 \text{ N}$
  - C  $490 \text{ N}$
  - D  $550 \text{ N}$
  - E  $590 \text{ N}$ .
6. Water flows at a rate of  $1.0 \times 10^6 \text{ kg}$  per second over the Victoria Falls.
- The Victoria Falls are  $120 \text{ m}$  high.
- The total power delivered by the water in falling through  $120 \text{ m}$  is:
- A  $1.2 \times 10^{12} \text{ W}$
  - B  $1.2 \times 10^9 \text{ W}$
  - C  $1.2 \times 10^8 \text{ W}$
  - D  $8.5 \times 10^{-10} \text{ W}$
  - E  $8.5 \times 10^{-11} \text{ W}$ .

[Turn over

7. A spacecraft passes the Earth at a speed of  $0.4c$ .

A light on the spacecraft pulses on and off.

A passenger on the spacecraft measures the time between the pulses as 2.5 s.

An observer on Earth measures the time between the pulses as:

- A 2.3 s
- B 2.5 s
- C 2.7 s
- D 3.0 s
- E 3.2 s.

8. A student makes the following statements about the expanding Universe:

- I The evidence supporting the existence of dark matter comes from estimations of the mass of galaxies.
- II The evidence supporting the existence of dark energy comes from the accelerating rate of expansion of the Universe.
- III The peak wavelength of radiation emitted by hotter stars is longer than that for cooler stars.

Which of these statements is/are correct?

- A I only
- B II only
- C III only
- D I and II only
- E I, II and III

9. A police car is travelling at a constant speed of  $31.0 \text{ m s}^{-1}$  towards a stationary observer. The siren on the car emits a sound with a frequency of 820 Hz.

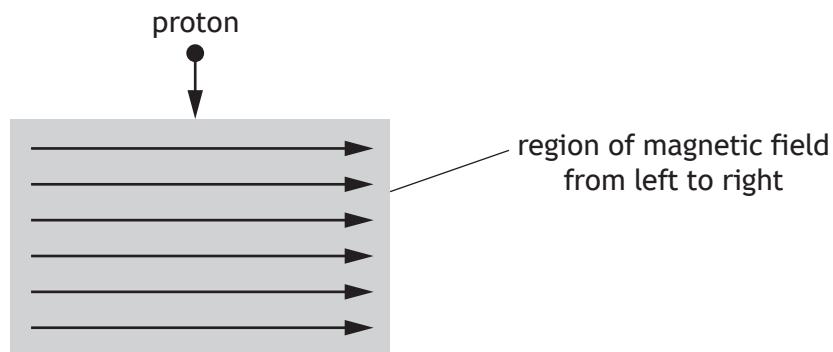
The speed of sound in air is  $340 \text{ m s}^{-1}$ .

The frequency of the sound heard by the observer is:

- A 745 Hz
- B 751 Hz
- C 820 Hz
- D 895 Hz
- E 902 Hz.



10. A proton enters a region of magnetic field as shown.



The direction of the force exerted by the magnetic field on the proton as it enters the field is:

- A out of the page
  - B into the page
  - C to the left
  - D to the right
  - E towards the bottom of the page.
11. The masses of three particles are shown.

| Particle    | Mass (kg)               |
|-------------|-------------------------|
| Electron    | $9.11 \times 10^{-31}$  |
| Proton      | $1.673 \times 10^{-27}$ |
| Higgs boson | $2.22 \times 10^{-25}$  |

How many orders of magnitude greater is the mass of a Higgs boson compared to the mass of a proton?

- A  $7.54 \times 10^{-3}$
- B 2
- C 5
- D 133
- E  $2.44 \times 10^5$

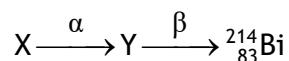
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12. A proton consists of two up quarks and a down quark.  
A student makes the following statements about protons:

- I Protons are baryons.
- II Protons are hadrons.
- III Protons are fermions.

Which of these statements is/are correct?

- A I only
  - B II only
  - C III only
  - D I and II only
  - E I, II and III
13. The following statement represents part of a radioactive decay series.



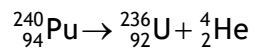
Nucleus X undergoes alpha emission to produce nucleus Y.

Nucleus Y then undergoes beta emission.

Nucleus X is:

- A  ${}^{218}_{85}\text{At}$
- B  ${}^{214}_{82}\text{Pb}$
- C  ${}^{218}_{84}\text{Po}$
- D  ${}^{218}_{86}\text{Rn}$
- E  ${}^{210}_{80}\text{Hg}$ .

14. The following statement represents a nuclear reaction.



The total mass of the particles before the reaction is  $398.626 \times 10^{-27} \text{ kg}$ .

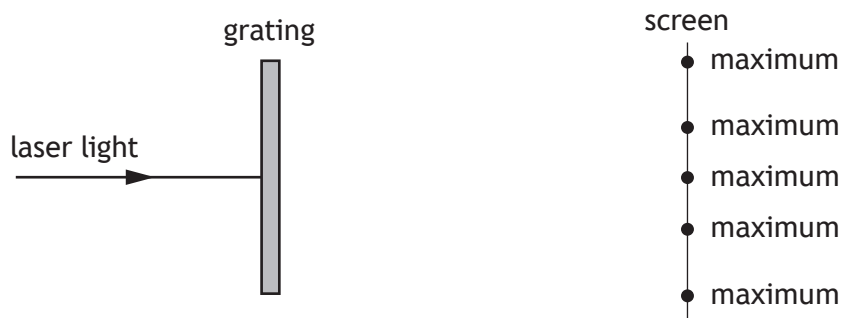
The total mass of the particles after the reaction is  $398.615 \times 10^{-27} \text{ kg}$ .

The energy released in this reaction is:

- A  $1.1 \times 10^{-29} \text{ J}$
  - B  $3.3 \times 10^{-21} \text{ J}$
  - C  $5.0 \times 10^{-13} \text{ J}$
  - D  $9.9 \times 10^{-13} \text{ J}$
  - E  $3.6 \times 10^{-8} \text{ J}$ .
15. The irradiance of light incident on a surface from a point source is  $20.0 \text{ W m}^{-2}$ .  
The distance between the point source and the surface is  $5.0 \text{ m}$ .  
The point source is now moved to a distance of  $25.0 \text{ m}$  from the surface.  
The irradiance of the light incident on the surface is now:
- A  $0.032 \text{ W m}^{-2}$
  - B  $0.80 \text{ W m}^{-2}$
  - C  $1.2 \text{ W m}^{-2}$
  - D  $4.0 \text{ W m}^{-2}$
  - E  $100 \text{ W m}^{-2}$ .

[Turn over

16. Light from a laser is incident on a grating as shown.



A series of interference maxima are observed on the screen.

A student makes the following statements about the interference pattern observed on the screen:

- I Increasing the distance between the grating and the screen increases the distance between the observed maxima.
- II Increasing the distance between the laser and the grating increases the distance between the observed maxima.
- III Decreasing the distance between the slits on the grating decreases the distance between the observed maxima.

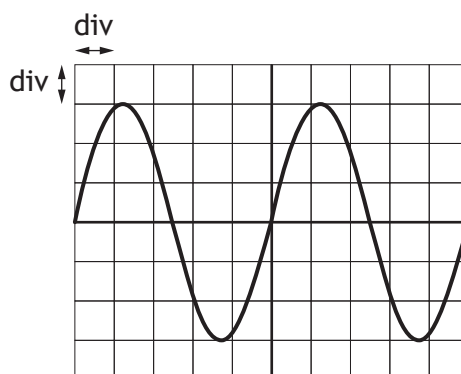
Which of the statements is/are correct?

- A I only
- B II only
- C I and III only
- D II and III only
- E I, II and III

17. Which row in the table shows what happens to the speed, frequency, and wavelength of red light as it passes from diamond into air?

|   | Speed     | Frequency | Wavelength |
|---|-----------|-----------|------------|
| A | decreases | decreases | no change  |
| B | decreases | no change | decreases  |
| C | decreases | increases | increases  |
| D | increases | no change | increases  |
| E | increases | increases | increases  |

18. The output from a signal generator is connected to an oscilloscope. The trace seen on the oscilloscope screen is shown.



The Y-gain setting on the oscilloscope is 2.0 V/div.

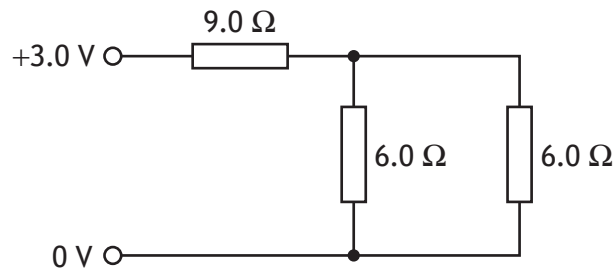
The time base setting on the oscilloscope is 5 ms/div.

Which row in the table gives the rms voltage and the frequency of the output from the signal generator?

|   | rms voltage (V) | Frequency (Hz) |
|---|-----------------|----------------|
| A | 4.2             | 25             |
| B | 4.2             | 40             |
| C | 6.0             | 40             |
| D | 6.0             | 200            |
| E | 8.5             | 25             |

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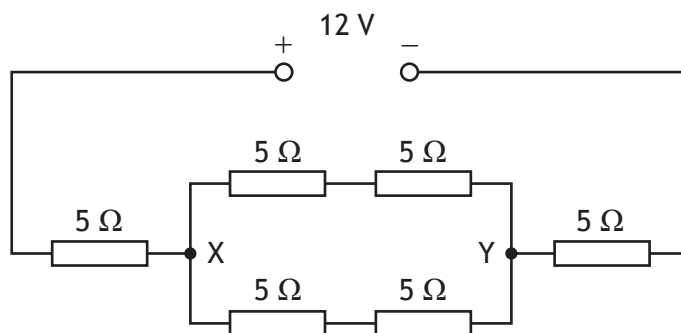
19. Three resistors are connected to a 3.0 V power supply as shown.



The power supply has negligible internal resistance.

The power dissipated in the circuit is:

- A 0.25 W
  - B 0.43 W
  - C 0.75 W
  - D 2.1 W
  - E 4.0 W.
20. Six resistors, each of resistance  $5\ \Omega$ , are connected to a 12 V power supply as shown.

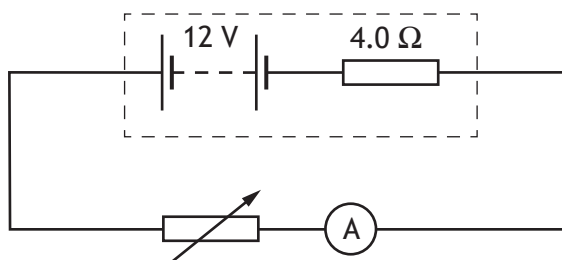


The power supply has negligible internal resistance.

Which row in the table shows the total circuit resistance and the potential difference across X and Y?

|   | Total circuit resistance ( $\Omega$ ) | Potential difference across X and Y (V) |
|---|---------------------------------------|-----------------------------------------|
| A | 15                                    | 2                                       |
| B | 15                                    | 4                                       |
| C | 20                                    | 6                                       |
| D | 30                                    | 8                                       |
| E | 30                                    | 12                                      |

21. A circuit is set up as shown.

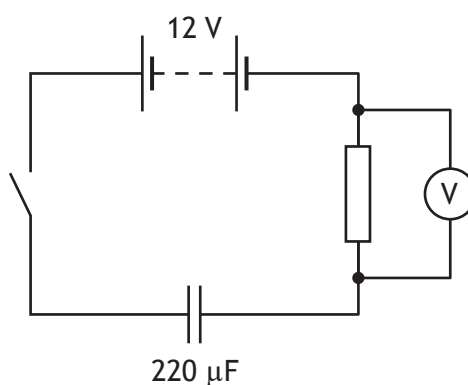


The resistance of the variable resistor is set to  $6.0\ \Omega$ .

The lost volts due to the internal resistance of the battery is:

- A  $1.2\ \text{V}$
- B  $4.8\ \text{V}$
- C  $6.0\ \text{V}$
- D  $7.2\ \text{V}$
- E  $8.0\ \text{V}$ .

22. A circuit is set up as shown.



The battery has negligible internal resistance.

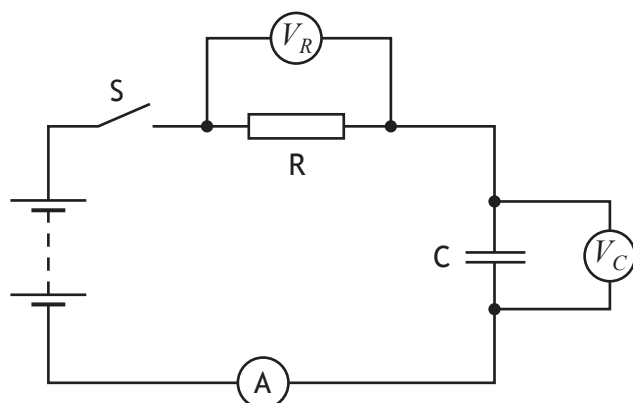
The capacitor is initially uncharged.

The switch is now closed.

When the reading on the voltmeter is  $7.0\ \text{V}$ , the charge stored on the capacitor is:

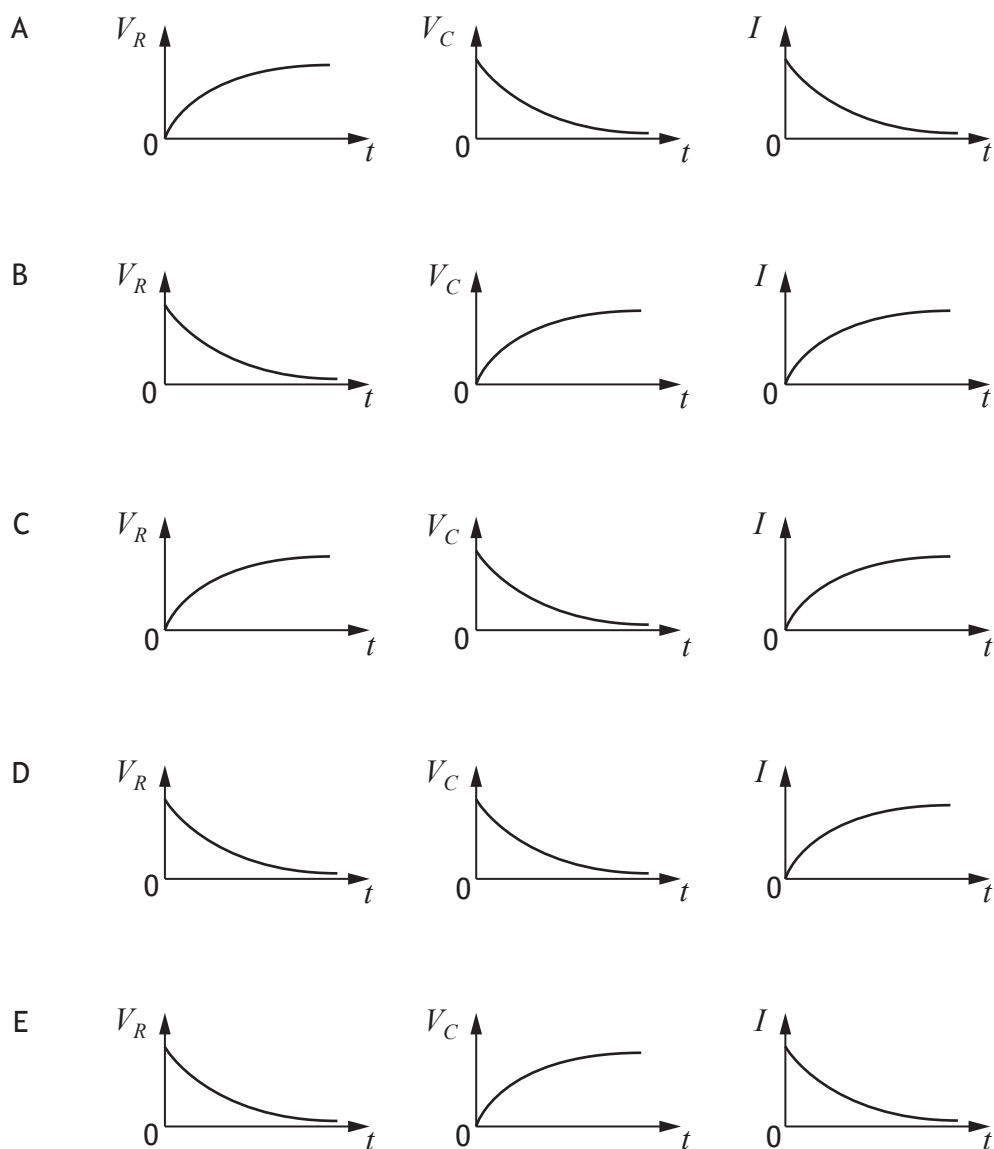
- A  $3.1 \times 10^{-5}\ \text{C}$
- B  $4.4 \times 10^{-5}\ \text{C}$
- C  $1.1 \times 10^{-3}\ \text{C}$
- D  $1.5 \times 10^{-3}\ \text{C}$
- E  $2.6 \times 10^{-3}\ \text{C}$ .

23. A circuit is set up as shown.



The capacitor is initially uncharged. Switch S is closed.

Which graphs show how the potential difference  $V_R$  across resistor R, the potential difference  $V_C$  across capacitor C, and the current  $I$  in the circuit, vary with time  $t$  as the capacitor charges?





24. Which row in the table describes the conduction band and the gap between the conduction band and the valence band in an insulator?

|   | Conduction band | Gap between conduction band and valence band |
|---|-----------------|----------------------------------------------|
| A | unfilled        | bands overlap                                |
| B | full            | bands overlap                                |
| C | unfilled        | large gap                                    |
| D | full            | small gap                                    |
| E | full            | large gap                                    |

25. Astronomers use the following relationship to estimate the mass  $M$  of a galaxy

$$M = \frac{v^2 r}{G}$$

where  $v$  is the orbital speed of a star in the outer regions of the galaxy, in  $\text{m s}^{-1}$

$r$  is the orbital radius of the star, in m

$G$  is the Universal Constant of Gravitation.

A star orbits at a radius of  $4.0 \times 10^{20}$  m in the outer regions of the Triangulum galaxy.

The orbital speed of the star is  $120 \text{ km s}^{-1}$ .

Based on this information, the mass of the Triangulum galaxy is:

- A  $3.8 \times 10^{20} \text{ kg}$
- B  $7.2 \times 10^{32} \text{ kg}$
- C  $8.6 \times 10^{34} \text{ kg}$
- D  $7.2 \times 10^{35} \text{ kg}$
- E  $8.6 \times 10^{40} \text{ kg}$ .

[END OF QUESTION PAPER]

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK

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